

1. Project Overview

For this project, I chose a craft brewery and taproom. The idea is to make a small database that can keep track of the beers the brewery offers, the ingredients used for each beer, production batches, and sales. Keeping this information in one place would make it easier to look up products and see basic sales and production information.

This is a sample database for a made-up brewery scenario. The beer names and example data are used to demonstrate how the database works; the prices, ingredient amounts, batch sizes, and sales records are not meant to represent actual brewery records.

2. How I Chose the Topic and Identified Requirements

I chose a craft brewery because it has several types of information that fit well into a relational database. A brewery needs to know what it makes, what goes into each beer, how much it produces, and what sells in the taproom.

For the first version, I identified the requirements by thinking through the basic tasks someone managing a small brewery would need to do. I focused on keeping the design manageable for this class project rather than trying to model every part of a real brewery.

The main requirements for the database are:

- Store each beer's name, style, alcohol percentage, and price.
- Keep a list of ingredients and their units of measurement.
- Connect beers to the ingredients used in their recipes.
- Record production batches, including date, amount produced, and status.
- Record taproom sales by beer, date, quantity, and total price.
- Allow basic queries that summarize sales and connect related information.

3. Database Design

The database is named `alis_craft_brewery`. It currently has five tables. Each table has a primary key, and foreign keys connect related records.

Table	What it stores
Beer	Beer name, style, ABV, and price.
Ingredient	Ingredient name, type, and unit of measurement.
Recipe	Connects a beer to an ingredient and stores the amount used.
Production_Batch	Beer being produced, batch date, gallons produced, and current status.

Sales	Beer sold, sale date and time, quantity, and total price.
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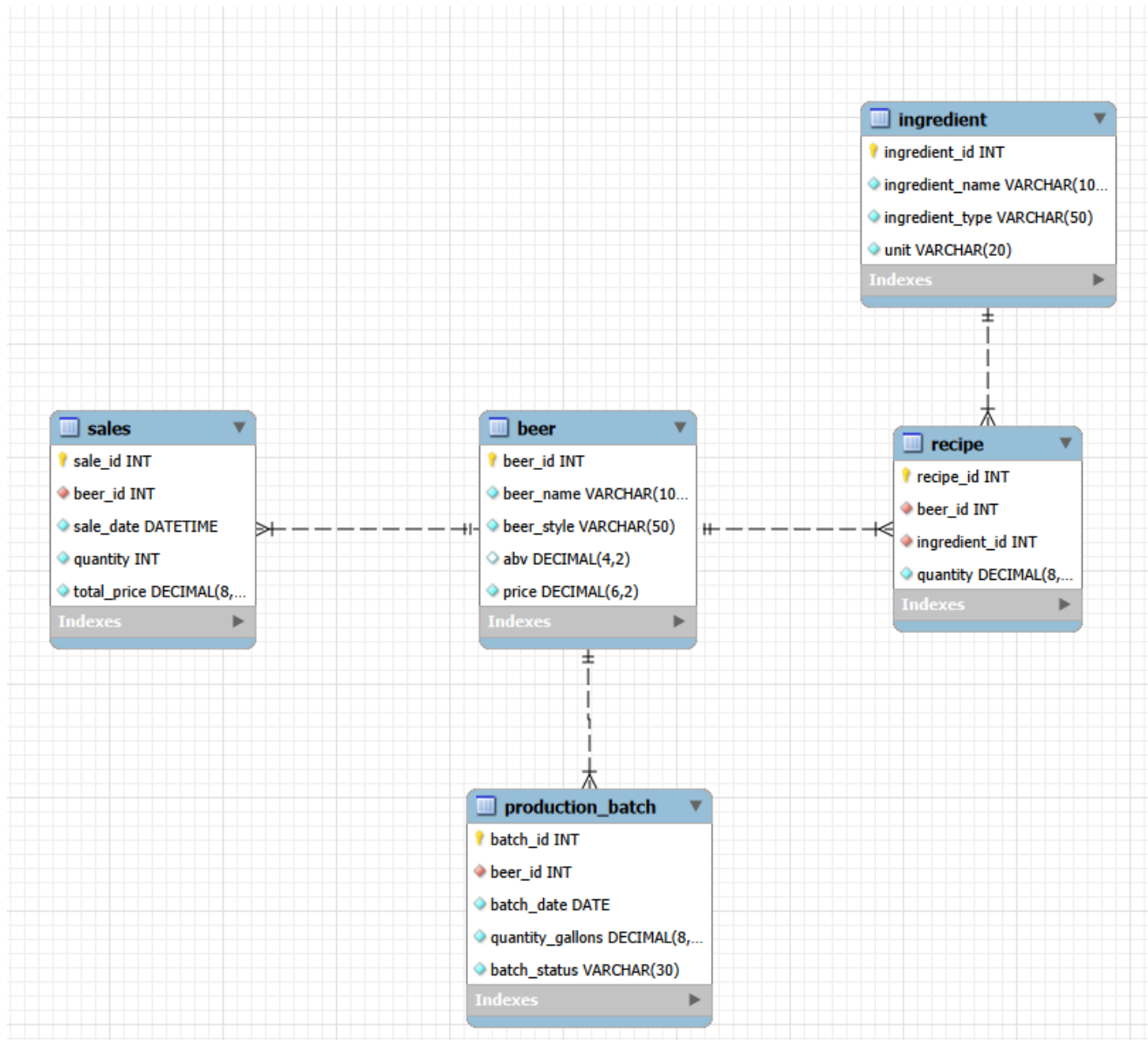
The relationship between Beer and Ingredient is many-to-many because one beer can use several ingredients, and one ingredient can be used in different beers. The Recipe table handles that relationship. A beer can also have multiple production batches and multiple sales records.

The Sales table is kept simple for this project: each row records a sale of one beer type. A larger system could separate a customer transaction from the individual items in that transaction, but that is outside the scope of this version.

4. ER Diagrams

Hand-drawn Chen ER diagram:

MySQL Workbench EER diagram:



5. Database Implementation

I created the database and tables in MySQL Workbench. The tables use primary keys to identify

records and foreign keys to connect related data. I also inserted sample records so I could test the relationships and run queries.

The sample data includes five beers, six ingredients, recipe records, production batches, and sales records. The sample values are only for testing the database.

The SQL script submitted with the project contains the commands to create the database and tables, insert the sample data, and run example queries.

6. Example Queries

One query joins Sales with Beer so the sales records show beer names instead of only beer IDs. Another query groups sales by beer and calculates the total quantity sold and total revenue for each beer.

Example query for total sales by beer:

```
SELECT Beer.beer_name, SUM(Sales.quantity) AS total_sold,  
SUM(Sales.total_price) AS total_revenue  
FROM Sales  
JOIN Beer ON Sales.beer_id = Beer.beer_id  
GROUP BY Beer.beer_name  
ORDER BY total_revenue DESC;
```

This query is useful because it gives a quick summary of which beers sold the most and how much revenue each one brought in, based on the sample records.

7. What I Learned

This project gave me practice turning a real-world scenario into tables and relationships. I also worked with primary keys, foreign keys, sample data, and SQL queries that join and summarize information. One thing I had to consider was how to keep the database simple enough for the assignment while still showing how the main parts of a brewery are connected.